

JS / KL Divergence Analysis

Epoch-wise Divergence from Baseline Distribution

Run Parameters

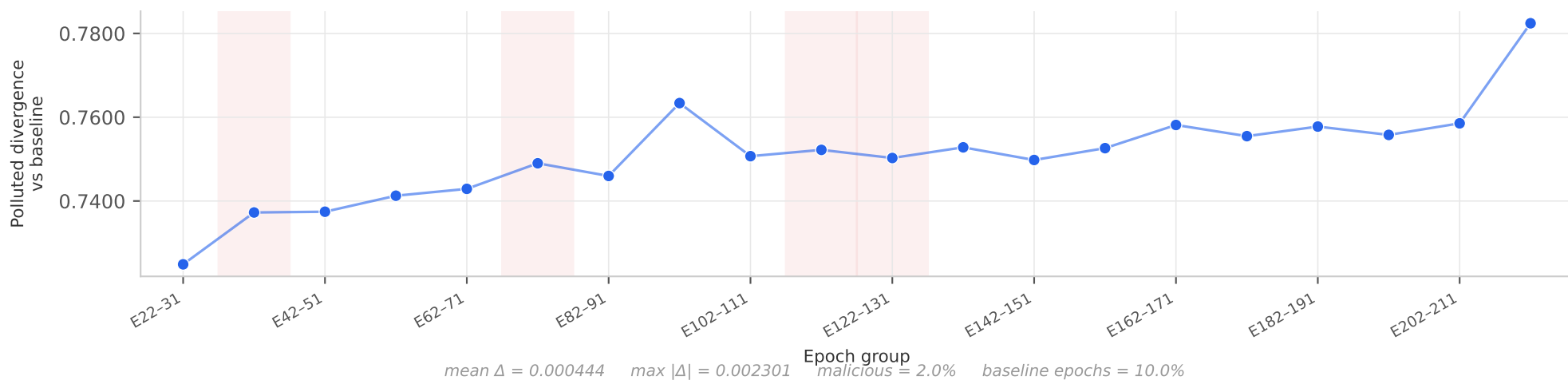
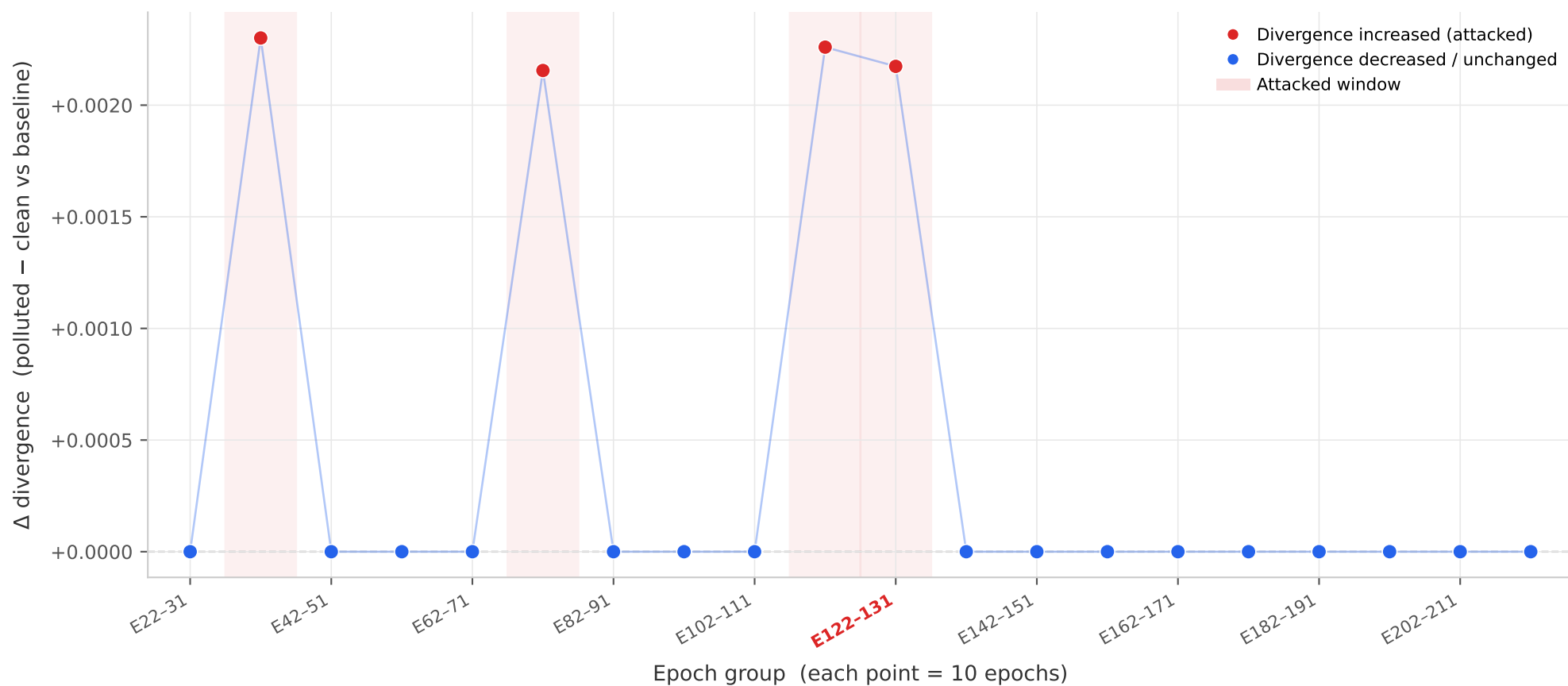
Dataset	epochs_10.0_2.0_sparse.json
Directory	CAIDA_Dataset/txtD280/epochs/
Timestamp	2026-02-19T19:21:04.817494
Method	Jensen-Shannon + KL
Total epochs	215
Baseline epochs	21 (10.0%)
Attack epochs	194
Malicious traffic	2.0%
Epoch group size	10 epochs per point
Metrics plotted	10
Mode	SPARSE
Attack probability	0.3
Windows total	20
Windows attacked	4
Windows clean	16

Baseline Distribution Summary

Total Flows	106645
Unique Src IPs	23780
Unique Dst IPs	16623
Unique Src Ports	14107
Unique Dst Ports	33214
Src IP Vocab Size	23780
Dst IP Vocab Size	16623
Src Port Vocab Size	14107
Dst Port Vocab Size	33214

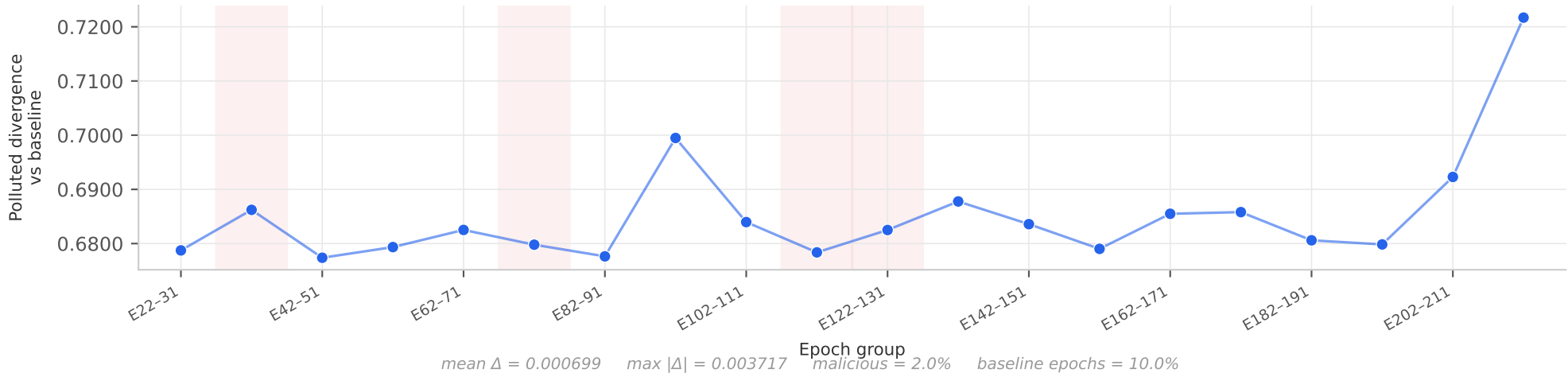
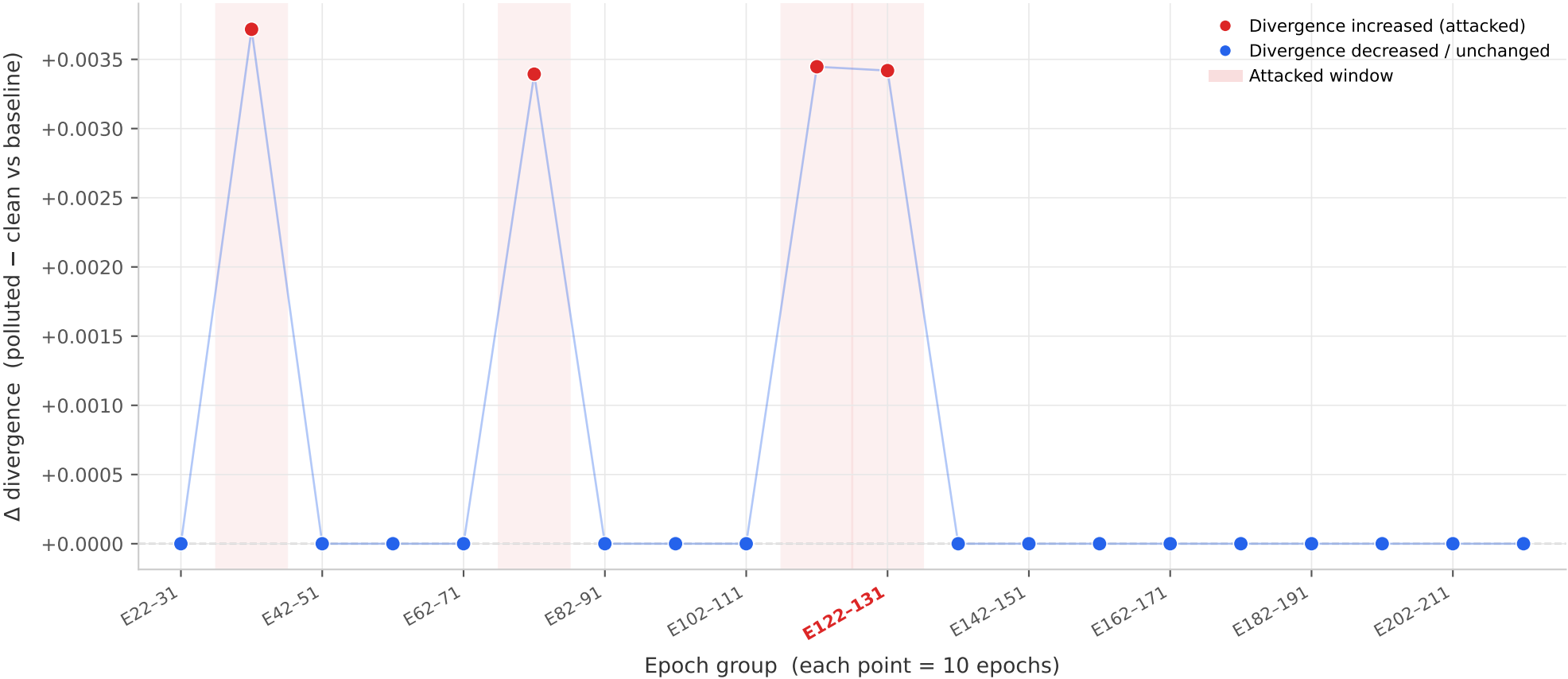
Source IP — JS Divergence

JS Divergence



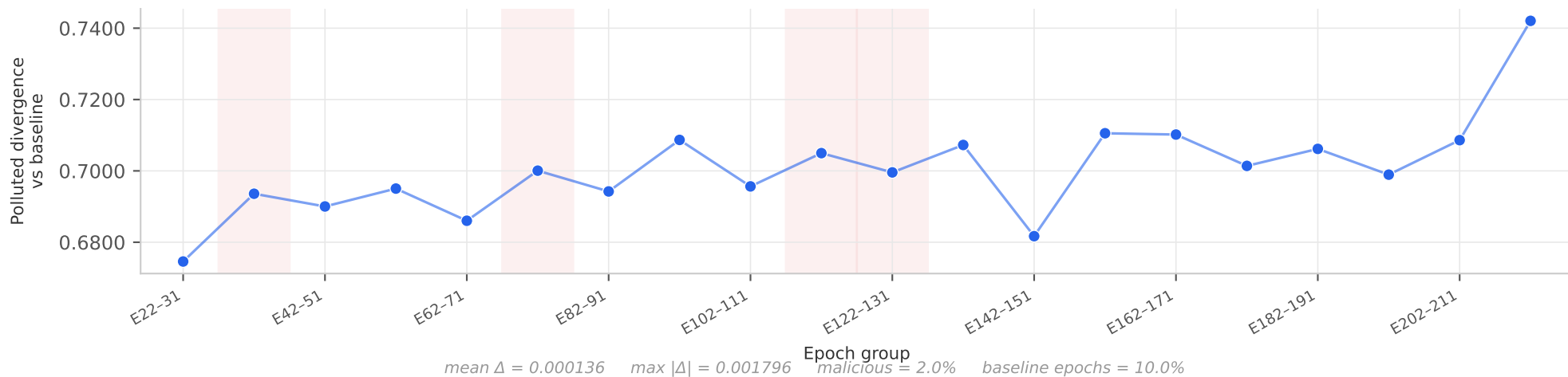
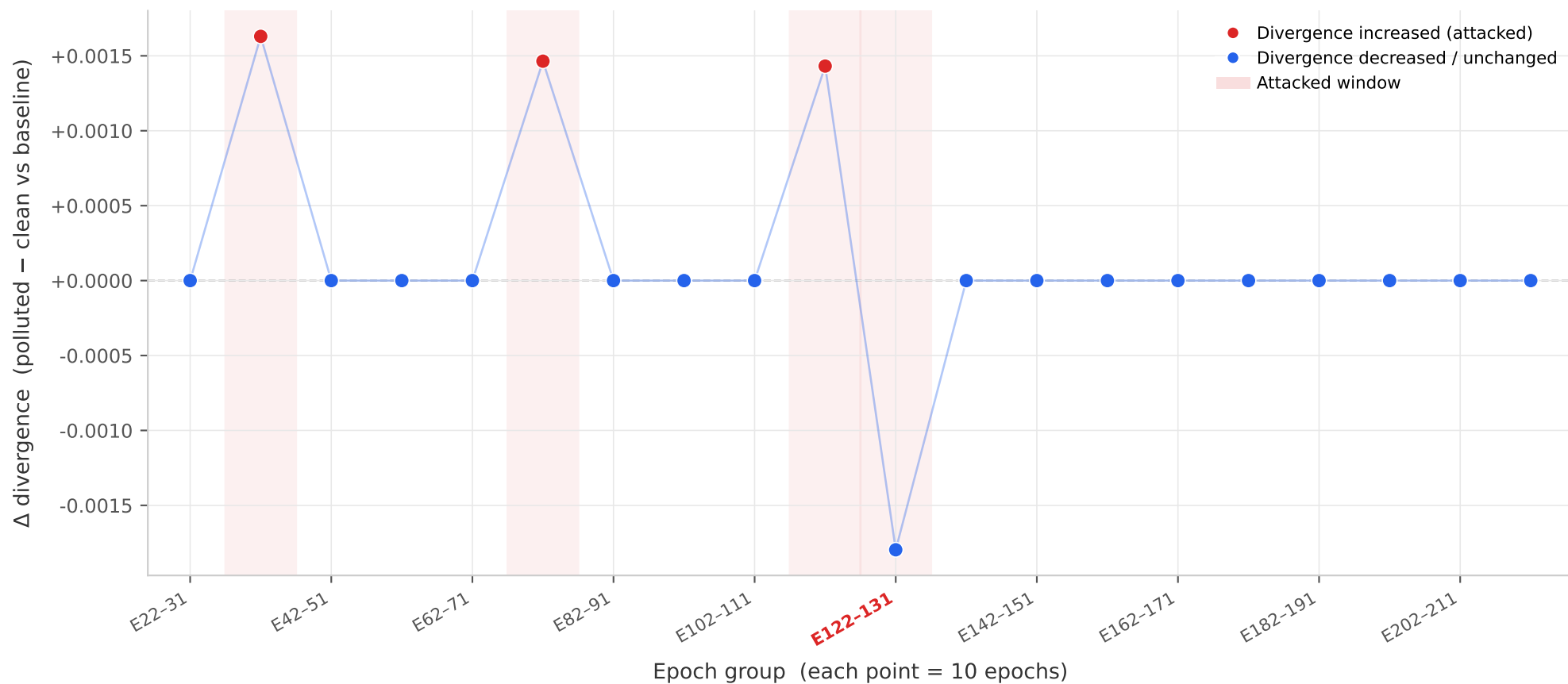
Destination IP — JS Divergence

JS Divergence



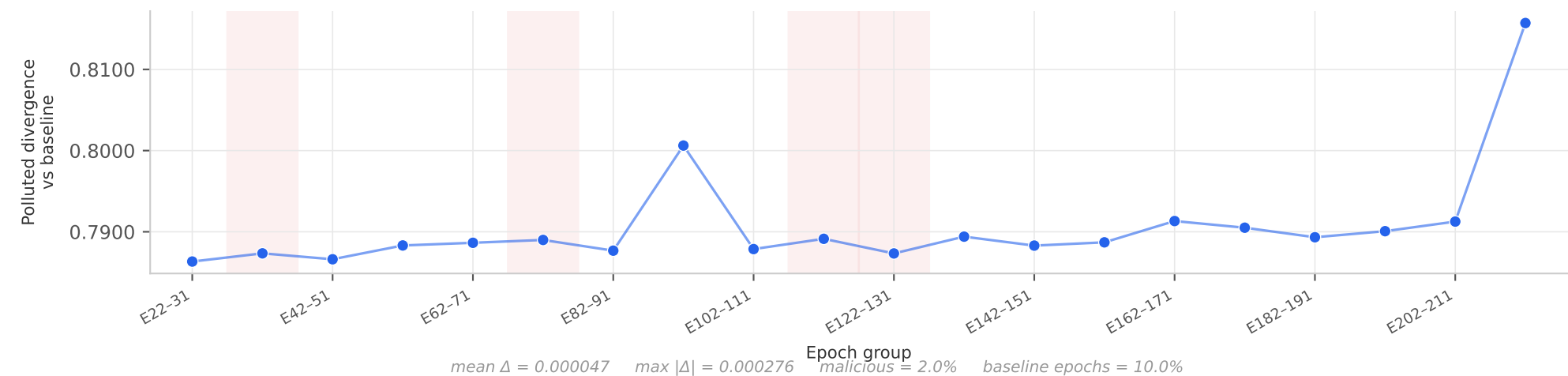
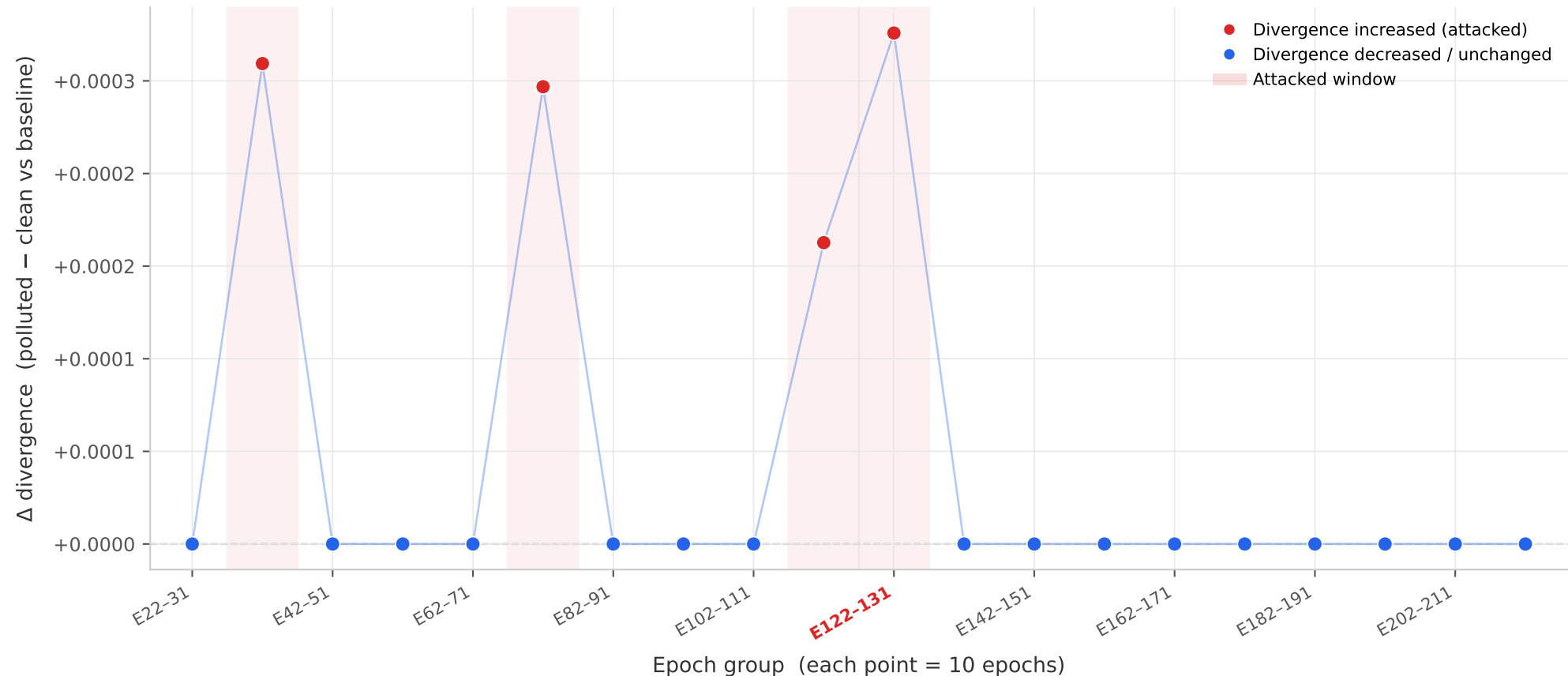
Source Port — JS Divergence

JS Divergence



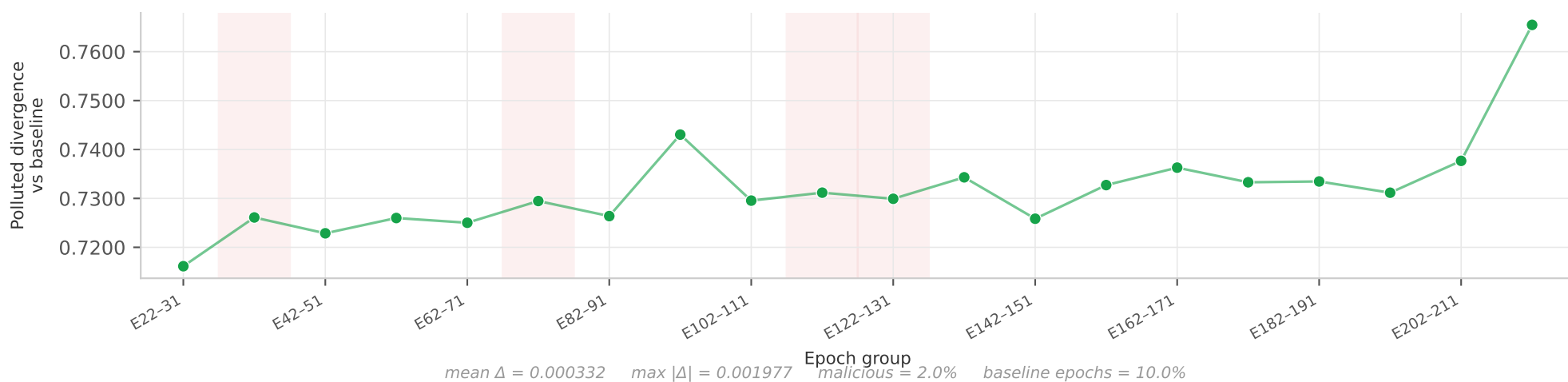
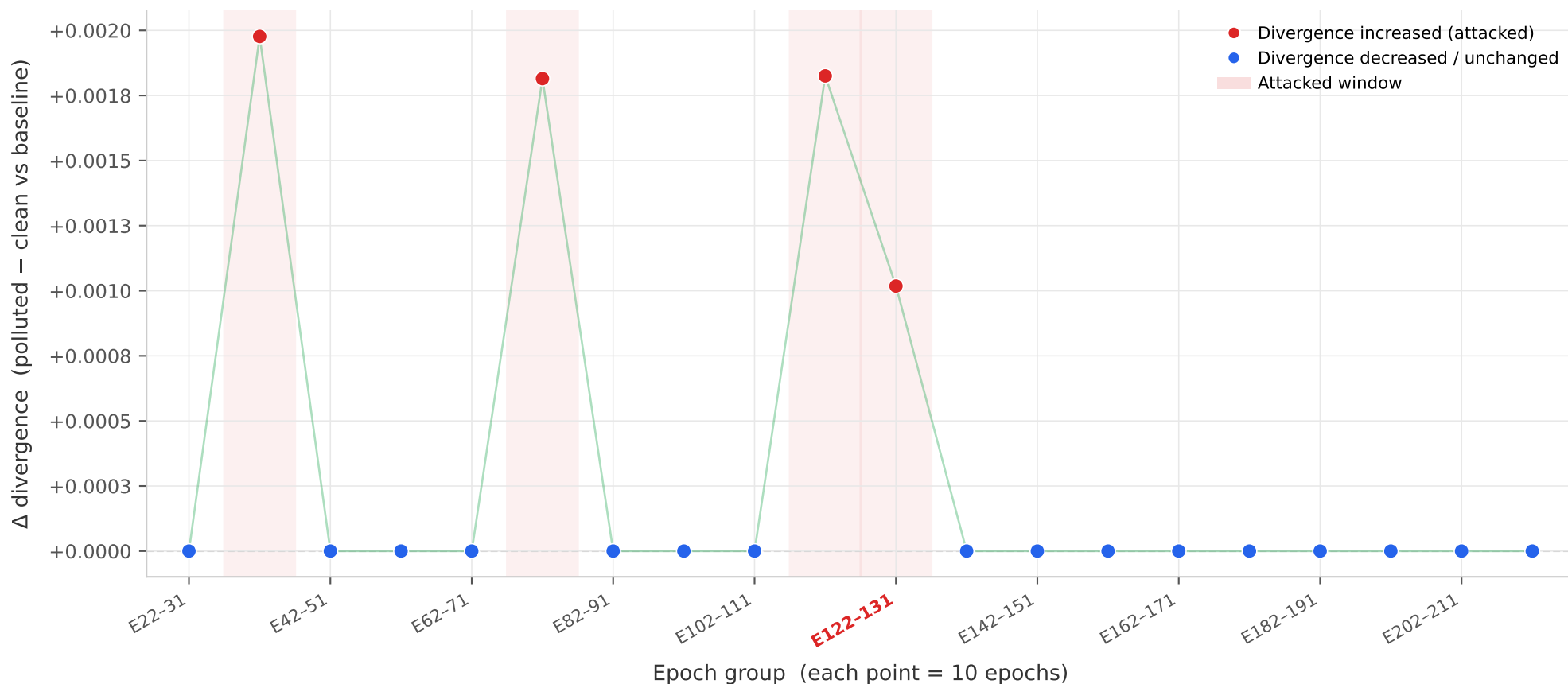
Destination Port — JS Divergence

JS Divergence



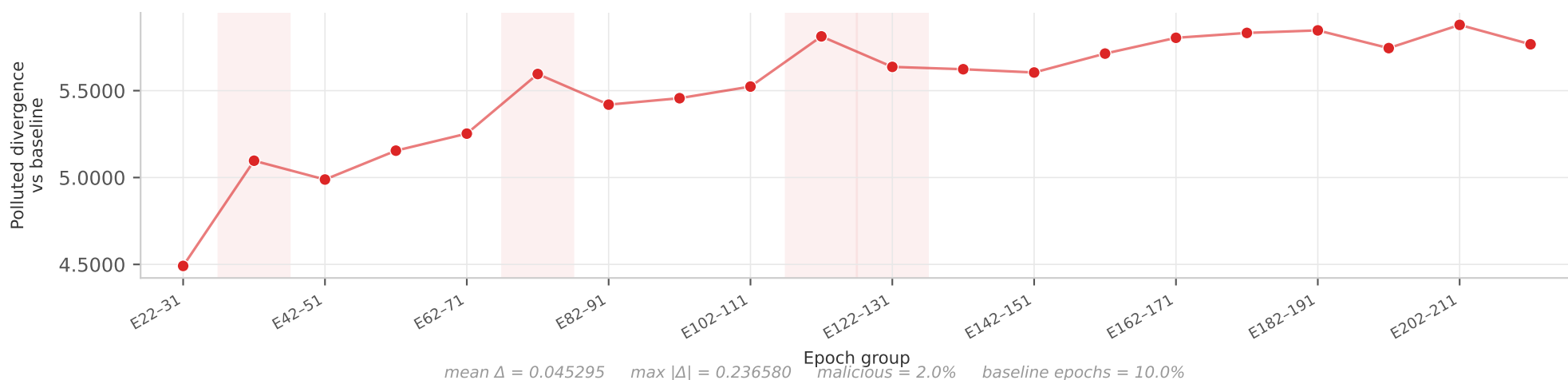
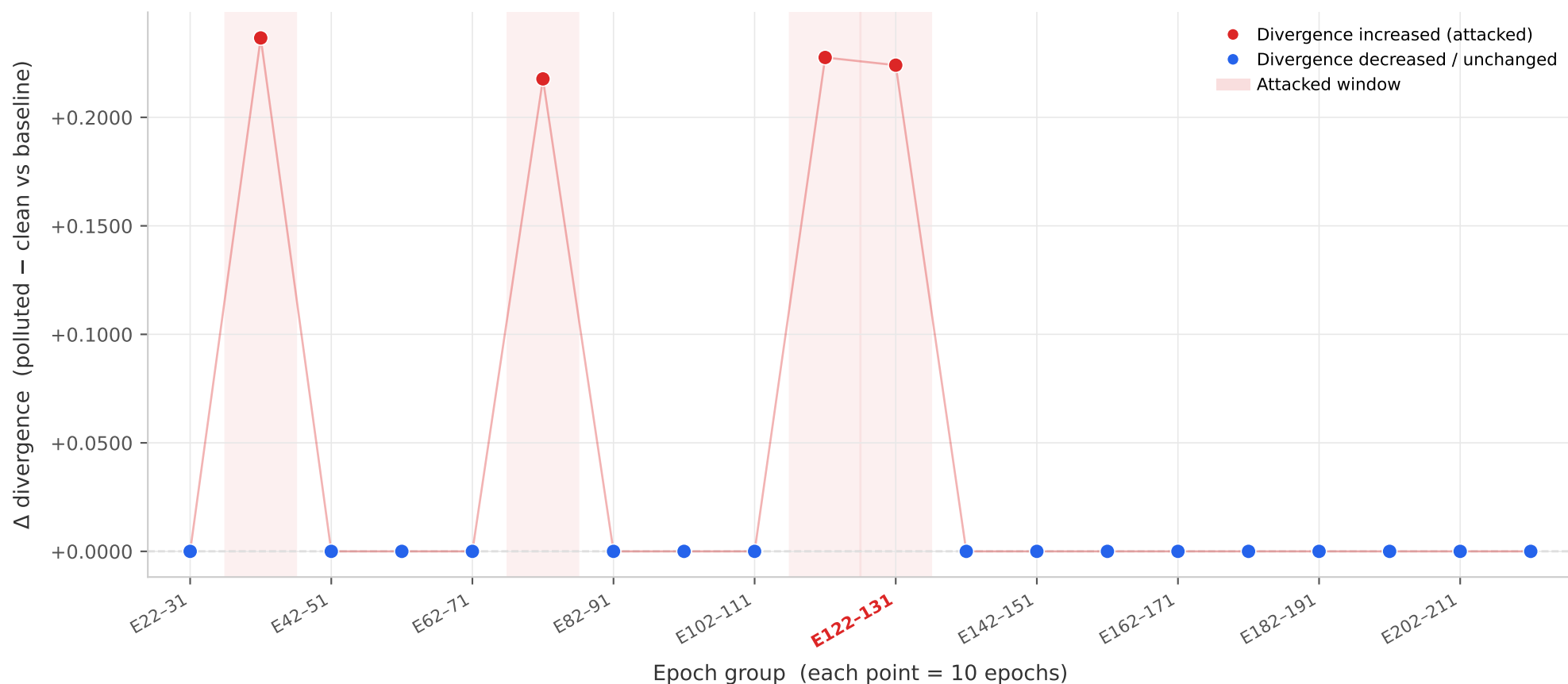
Combined (all features) — JS

Combined



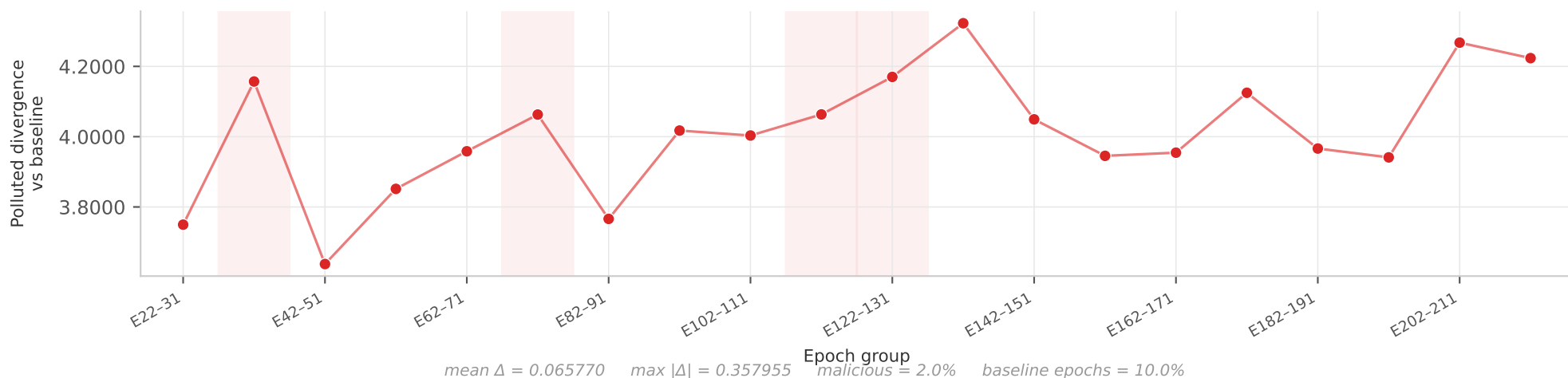
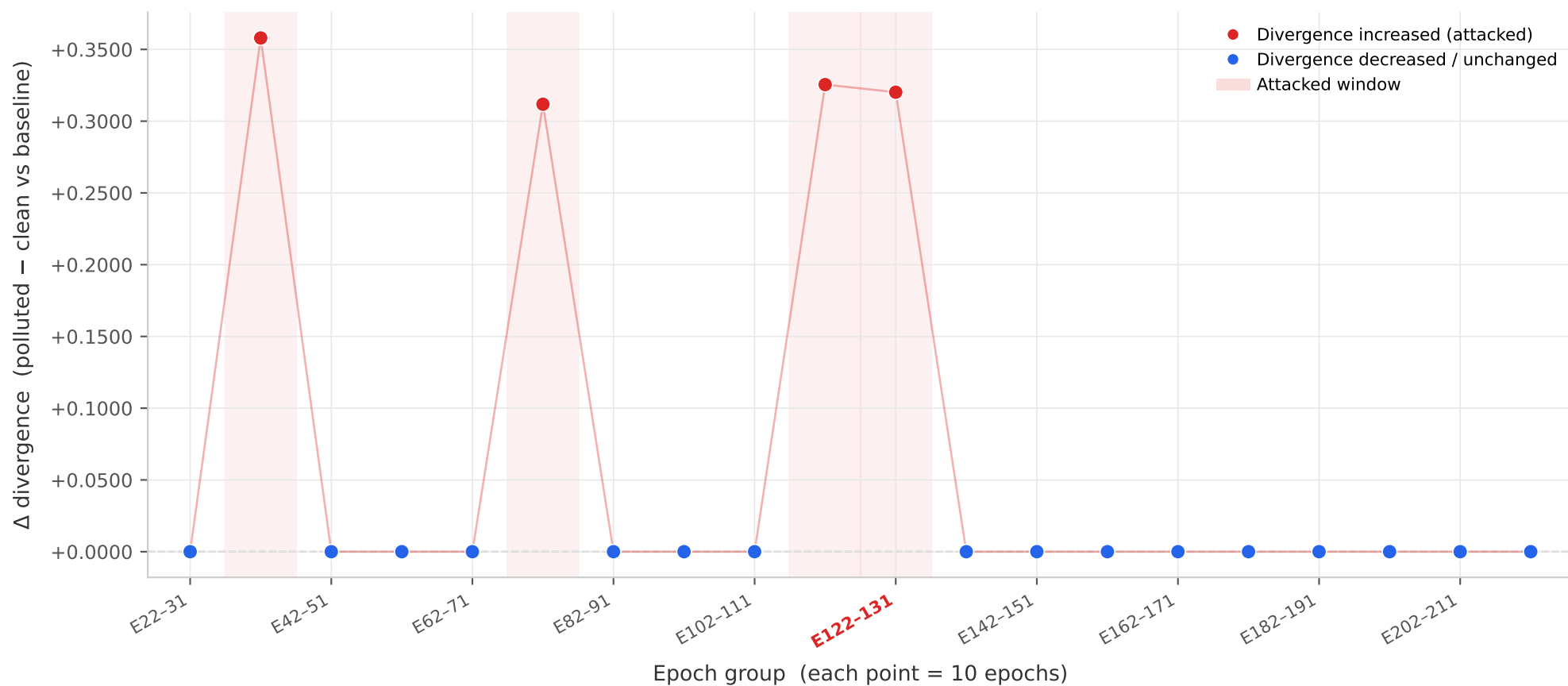
Source IP — KL Divergence

KL Divergence



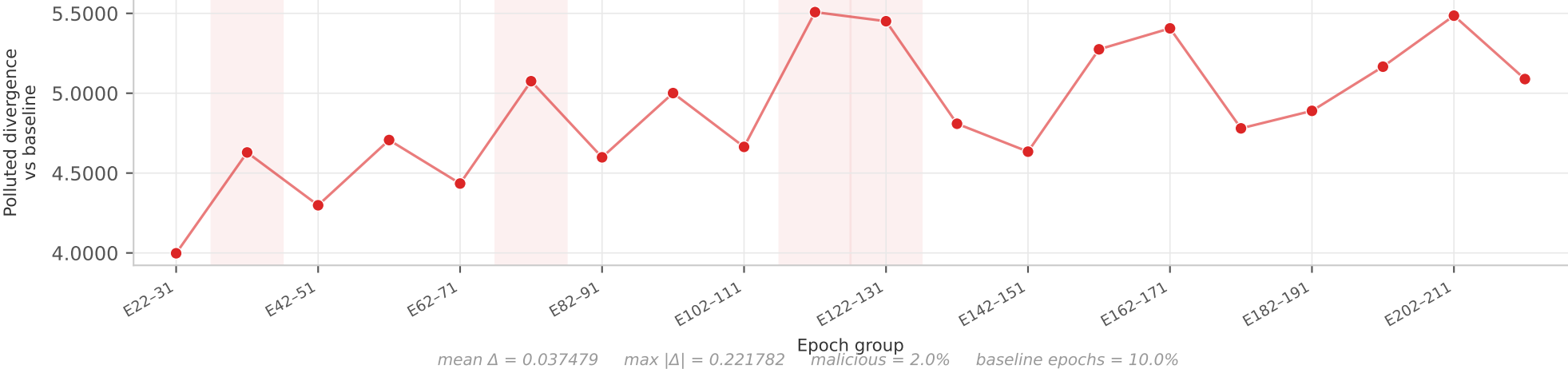
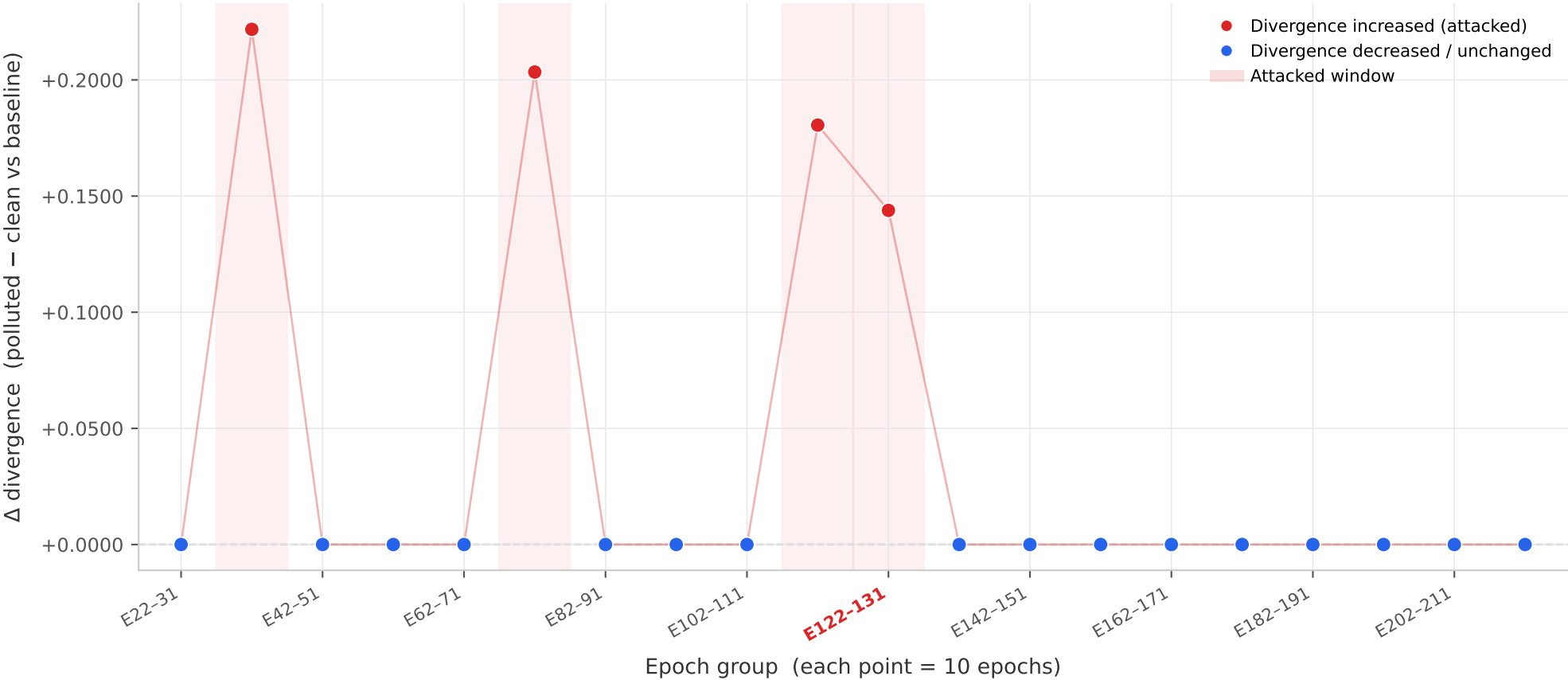
Destination IP — KL Divergence

KL Divergence



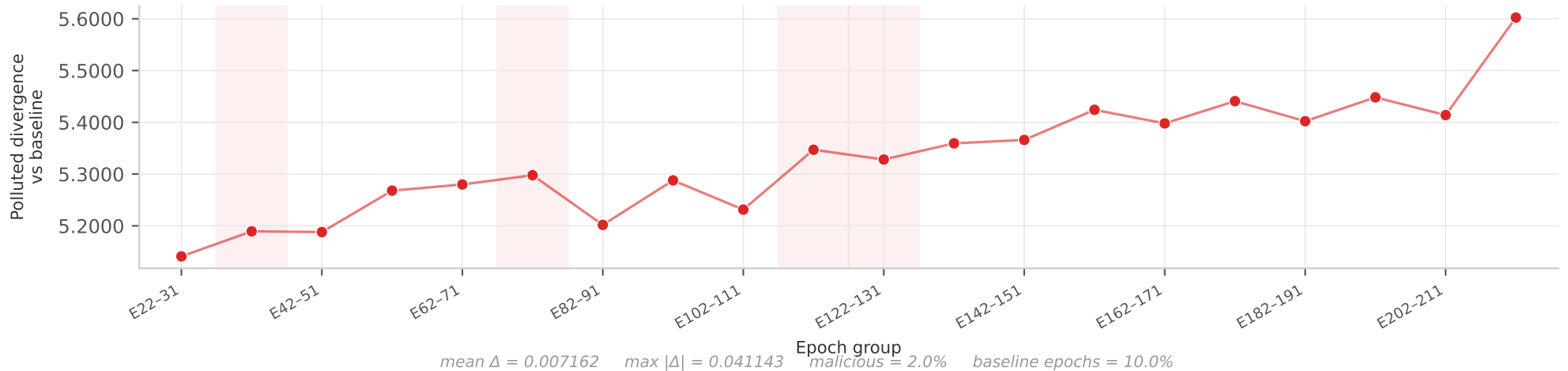
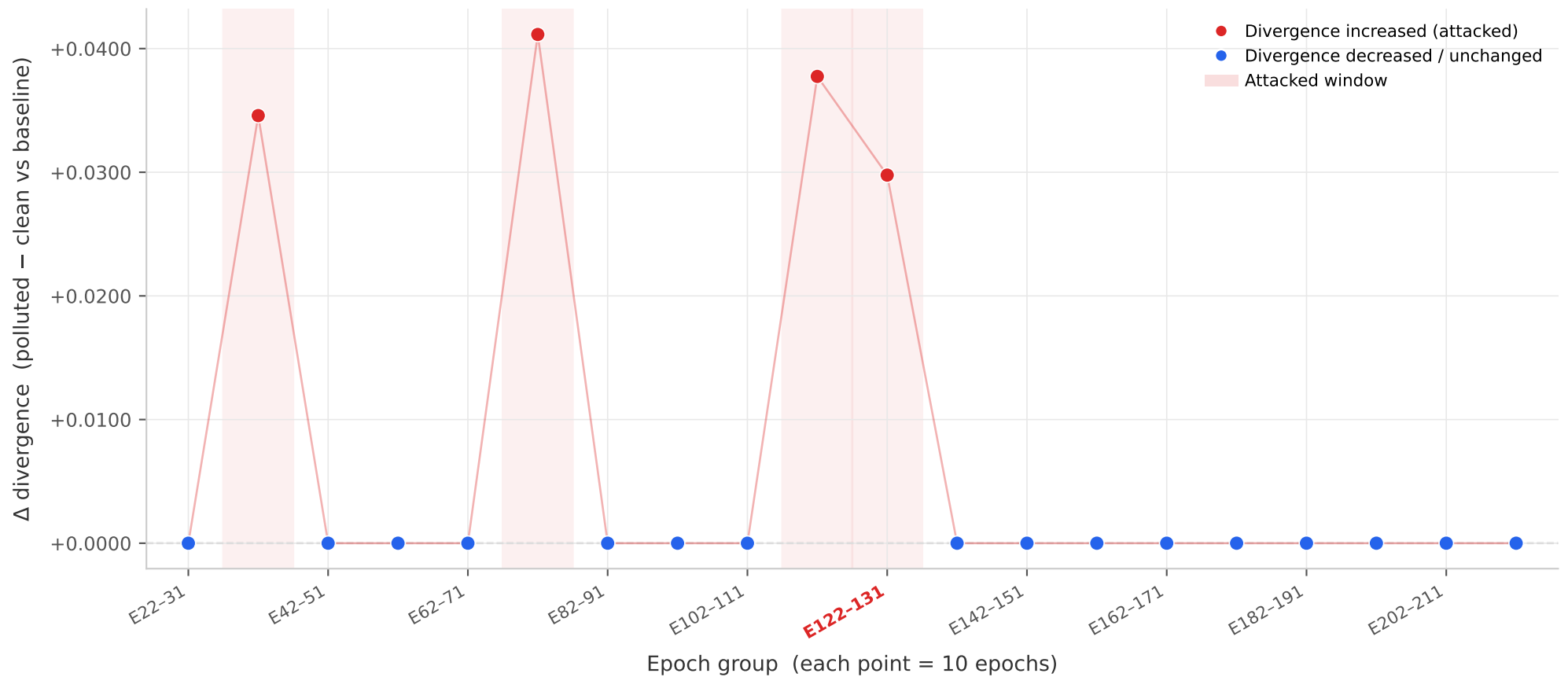
Source Port — KL Divergence

KL Divergence



Destination Port — KL Divergence

KL Divergence



Total Flows

Flow

